Reconstructing PG(2, 13), Its Conic, and Polarity from the Minimum Words of a Binary Conic Code

Tavis Rudd

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Abstract

Let K be the binary nullspace of the passant-by-internal incidence matrix of a nonsingular conic in $\mathrm{PG}(2, 13)$. We prove that K has parameters $[78, 36, 12]_2$ and exactly 364 minimum words, forming four $\mathrm{PGL}(2, 13)$-orbits—one octahedral family and three chord-indexed punctured-conic families—each spanning K.

From weighted pair concurrences in the minimum supports we recover the 78 passant incidence rows and the six-class elliptic association scheme; pair parity alone recovers K. The recovered scheme has automorphism group $\mathrm{PGL}(2, 13)$, whose Sylow-13 subgroups and involutions let us reconstruct all 183 points and lines of $\mathrm{PG}(2, 13)$, the distinguished conic, and its polarity, without coordinates or a projective frame. The weighted 2-section of the minimum-support hypergraph is therefore a complete invariant of this marked presentation up to isomorphism, while unary minimum-support statistics are constant: reconstruction has exact arity two. The binary relation algebra acts on K through $\mathbb{F}_8$, making K twelve-dimensional over that field.

The distance proof excludes weight eight by a positive-semidefinite clique obstruction and weight ten by two line-moment profiles followed by bounded stabilizer exhaustion. The theorem is intentionally specific to $q = 13$: it gives complete ambient reconstruction, not a uniform distance formula.

Keywords. binary conic code; passant line; minimum-weight codeword; weighted hypergraph 2-section; projective-plane reconstruction; elliptic association scheme; polarity; automorphism group.

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1 Introduction and statement

We ask how little of an incidence code is needed to reconstruct the geometry that produced it. This is a fixed-field inverse problem: starting from the minimum-support hypergraph, we ask for the least arity of support data needed to reconstruct the geometry erased by row reduction. The retained input is only the weighted pair data inside the minimum supports of a binary code; neither its parity-check matrix nor an ambient projective plane is supplied. At $q = 13$ this pairwise shadow is complete: it recovers the incidence matrix, the elliptic association scheme, the full automorphism group, and finally the marked conic plane and its polarity. Unary data there are constant, so the exact reconstruction arity is two.

Fix a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2, 13)$. A line is *passant* when it is disjoint from $\mathcal{C}(\mathbb{F}_{13})$. Conic polarity identifies the 78 internal points with the 78 passant lines. Let M be their binary incidence matrix and put

$$K = \ker_{\mathbb{F}_2} M \subseteq \mathbb{F}_2^{78}.$$

Write $G_0 = \mathrm{PGL}(2,13)$ for the conic stabilizer in its natural action. The coordinate set is the set of internal points. Automorphisms below are coordinate permutations preserving the indicated code or minimum-support system.

Let

$$\mathcal{H} = \{\mathrm{supp}(c) : c \in K,\ \mathrm{wt}(c) = 12\}$$

be the minimum-support hypergraph on the unlabeled coordinate set of K.

Theorem 1.1 (Minimum words recover the conic plane)**.** *The code K has parameters $[78, 36, 12]_2$ and exactly 364 minimum words. Under $\mathrm{PGL}(2,13)$, those words form four orbits of size 91. One is an octahedral homogeneous space with stabilizer S_4; the other three are chord-indexed punctured conics with stabilizer D_{24}. Every orbit spans K.*

Let $c(P,Q)$ be the number of minimum supports containing P, Q. The neighborhoods defined by $c(P,Q) = 8$ are exactly the 78 rows of M. The other five elliptic relations are recovered from the remaining pair weights and one common-neighbor count. Moreover, if $(R_{\mathcal{H}})_{PQ} = c(P,Q) \bmod 2$ off the diagonal and $(R_{\mathcal{H}})_{PP} = 0$, then $\operatorname{im} R_{\mathcal{H}} = K$. Thus the weighted 2-section of $\mathcal{H}$ reconstructs the incidence matrix, code, and elliptic scheme.

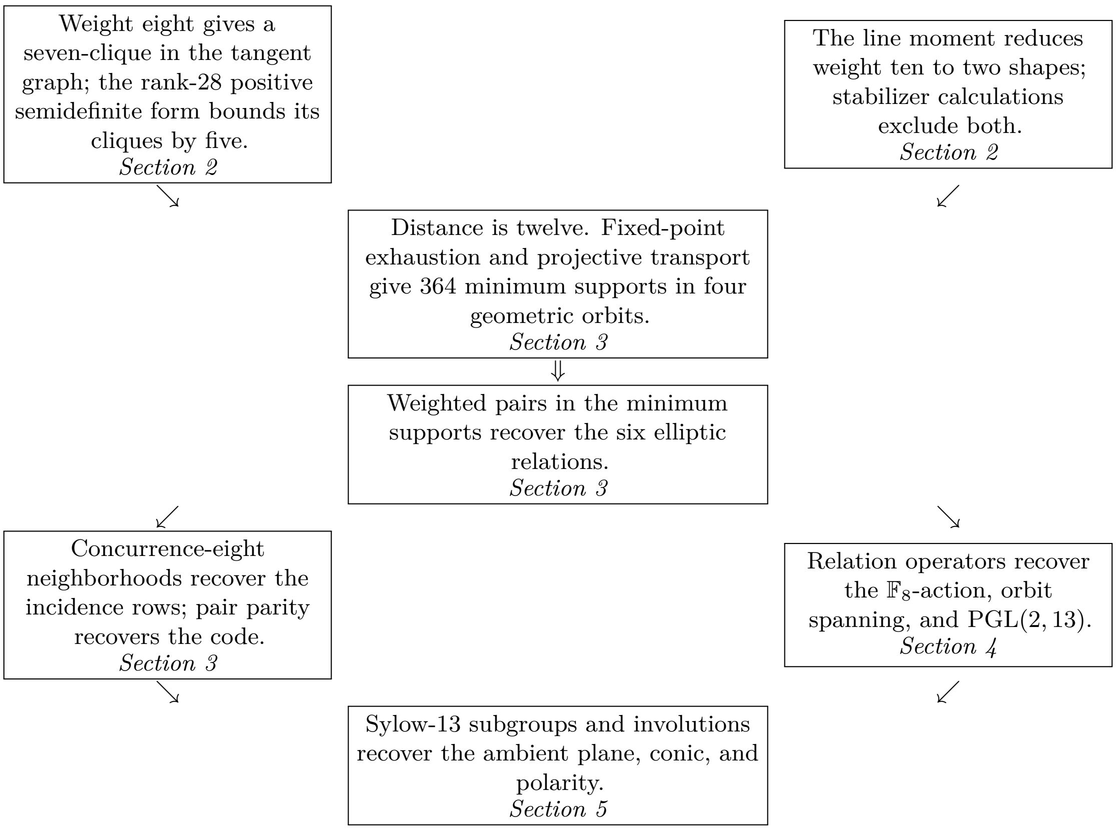
From the reconstructed group, its fourteen Sylow-13 subgroups and 169 involutions recover all 183 points and lines of $\mathrm{PG}(2,13)$, the distinguished conic, and its polarity. Consequently weighted-pair reconstruction has exact arity two, and

$$\mathrm{Aut}(K) = \mathrm{Aut}(\mathcal{H}) = \mathrm{PGL}(2,13)$$

in its symmetric-square action on the 78 internal points.

Reconstruction is intrinsic, not coordinatized. It recovers the marked plane and polarity up to isomorphism, but it does not choose an ordered projective frame, a field labeling, or an equation for the conic. Such a choice is necessarily noncanonical: the 2184 ordered triples of distinct conic points form a torsor for $\mathrm{PGL}(2,13)$.

Proof and reading map.



The reductions are structural except at small exact endpoints. The finite leaves are the four line-stabilizer and thirty-three point-stabilizer orbits in weight ten, the one-point minimum-layer exhaustion, and one bounded matching-profile check for the octahedral construction. The recovered group/plane dictionary is independently replayed in the complete finite model.

The code belongs to the fixed-conic LDPC family introduced by Droms–Mellinger–Meyer [2]. Their general bounds specialize here to $8 \leq d \leq 12$ [2, Theorem 4.9], and they report the $q = 13$ parameter $[78, 36, 12]_2$ in their Table 8. The later comparison table of Ma–Liu–Tian records the same general interval [6, Table 1]. Madison and Wu proved the general nullity formula $(q-1)^2/4$ [5, Corollary 5.4]; their module theorem [5, Theorem 5.1(i)] splits the code over $\overline{\mathbb{F}_2}$ into three non-isomorphic simple twelve-dimensional $\mathrm{PSL}(2,13)$-modules. In Section 4 we mark them by the eigenvalues $\alpha, \alpha^2, \alpha^4$ of A_9. Frobenius cycles the three constituents, so their binary descent is irreducible; Schur descent then gives endomorphism field $\mathbb{F}_8$. The unmarked irreducibility already forces any nonzero minimum-word orbit to span. We mark A_9 among the three conjugates, recover it from pair data, and give direct Gram certificates for each of the four orbits.

Hollmann and Xiang constructed the elliptic association scheme afforded by the same conic action [4, Sections 3–5], attributing its first description to Hollmann's thesis [3]. The identification of involutions with off-conic points, with involution axis equal to polar line, is classical and is stated explicitly by Tranchida [7, Section 2]. We prove the reported distance at $q = 13$, classify all minimum words, and prove that their weighted pair data recovers these established ambient structures.

2 Distance twelve

We use the model $\mathcal{C} : XZ - Y^2 = 0$. Every row and column of M has weight seven, and conic polarity identifies the two index sets. The Madison–Wu formula gives $\dim K = 36$; direct binary elimination is an independent check of rank 42.

Summing all row equations gives the all-one functional, so every word of K has even weight. If a nonzero word contains an internal point P, each of the seven passants through P contains a second support point. Those seven points are distinct. Thus every nonzero word has weight at least eight.

2.1 The tangent obstruction in weight eight

The mechanism has two steps. A saturated passant pencil turns a hypothetical weight-eight support into a seven-clique in a local graph; a positive-semidefinite form bounds every clique in that graph by five. On a first reading, the matrix entries below may be skipped once these two conclusions and the exact rank certificate are retained.

Suppose that a word has weight eight and fix a support point P. Each of the seven passants through P must contain another support point. Since only seven points remain, each passant contains exactly one of them. Applying the same argument at every support point shows that every support join is passant and that no three support points are collinear. Thus the support is an eight-arc, and every passant pencil at a support point is saturated. The seven remaining lines through each support point are therefore both the combinatorial tangents to the arc and the conic secants through that point.

For each secant line choose a nonzero linear equation. For an internal point P, put

$$T_P(X) = \prod_{\substack{\ell \ni P \\ \ell \text{ secant to } \mathcal{C}}} \ell(X).$$

For a pairwise-passant triple define

$$h(P,Q,R)=\frac{T_P(Q)T_Q(R)T_R(P)}{T_P(R)T_R(Q)T_Q(P)}.$$

Every displayed evaluation is nonzero because the three joins are passant. Rescaling a line equation multiplies one numerator and one denominator factor by the same scalar, so h is well defined. The saturated-pencil identification above makes the factors of T_P precisely the combinatorial tangents at P. Segre's lemma of tangents in the coordinate-free form of Ball–Lavrauw [1, Lemma 27] gives $h(P,Q,R)=1$ for every triple in the hypothetical support. Its hypotheses apply with $t=13+3-1-8=7$; hence the sign $(-1)^{t+1}$ is $+1$. With the cyclic order P,Q,R its tangent-product quotient is exactly the displayed h (not its negative; reversing the cycle only inverts it). Fix $P=(1:0:2)$. Its other seven points would form a clique in the graph on the 42 passant-join neighbors of P, where Q and R are adjacent when QR is passant and $h(P,Q,R)=1$.

The projective map

$$\gamma(x:y:z)=(x+6y+9z:7x+9y+3z:10x+y+z)$$

fixes P and has order 14 on its passant-join neighbors. With indices in $\mathbf{Z}/14$, put

$$A_i=\gamma^i(1:1:7),\qquad B_i=\gamma^i(1:1:6),\qquad C_i=\gamma^i(1:2:10).$$

These three orbits are disjoint and contain all 42 neighbors. Direct substitution in the product above gives

orbit pair	$j-i \pmod{14}$
AA	$4,6,8,10$
AB	$6,7,11,12$
AC	$1,3$
BB	$6,8$
BC	$3,5,6,8,9,11$
CC	$2,4,10,12.$

Let C be the symmetric 42-square matrix in the displayed vertex order, written in 14-square circulant blocks. Its six upper-block first rows are

00	40	9	8	−8	−10	0	−10	−2	−10	0	−10	−8	8	9
01	7	6	6	6	7	−1	−10	−10	−8	0	−8	−10	−10	−1
02	0	−10	2	−10	0	3	2	9	2	4	2	9	2	3
11	40	15	10	10	−6	−3	−10	−22	−10	−3	−6	10	10	15
12	−13	3	0	−10	8	−10	−10	14	−10	−10	8	−10	0	3
22	40	−9	−10	19	−10	2	8	−12	8	2	−10	19	−10	−9.

The lower blocks are the transposes. Comparison with the six difference sets shows that C has diagonal 40 and entry -10 on every edge of the local graph.

The matrix is positive semidefinite. Exact cyclic Fourier traces and Newton identities give

$$\chi_C(x)=x^{14}(x^2-94x+279)(x^2-26x+135)f(x)^2g(x)^2,$$

where

$$f(x)=x^6-247x^5+22466x^4-910807x^3+15948097x^2-111043270x+189747571,$$
$$g(x)=x^6-533x^5+105578x^4-9736765x^3+438572569x^2-9062718662x+68467048091.$$

Every nonzero factor has alternating coefficients. Its value at $-t$ is therefore positive for $t \geq 0$. Since C is real symmetric, it has no negative eigenvalue; the displayed factorization also gives rank 28. An independent rational Schur elimination gives the same twenty-eight positive pivots.

If v is the characteristic vector of a clique of size $c > 0$, then

$$0 \leq v^{\mathsf{T}} C v = 40c - 20\binom{c}{2} = 10c(5-c).$$

Thus $c \leq 5$. The set $\{A_0, B_6, B_{12}, C_1, C_3\}$ is a five-clique, so the bound is sharp. The exact certificate also classifies the extremal five-cliques. This sharpening is not used to exclude weight eight. Its fourteen translates are linearly independent kernel vectors; nullity fourteen makes them a basis of $\ker C$. Equality in the quadratic-form bound puts every maximum-clique vector in this kernel, and solving the resulting 0–1 constraints in that basis gives precisely the fourteen translates. In particular the seven-clique required by a weight-eight word cannot occur.

2.2 The two weight-ten profiles

Here the conceptual step is the line moment, which reduces every possible support to two geometric shapes. The stabilizer calculations that follow close those two finite leaves and introduce no further shape.

Let S be a hypothetical weight-ten support, and let n_i be the number of passants meeting S in exactly i points. Code parity makes every occurring i even, and incidence counting gives

$$\sum_i i n_i = 70.$$

At a selected point, pencil parity gives secant degree zero or two. The secant graph induced on S is therefore a union of cycles and isolated vertices. If m is the number of its cycle vertices, then it has m edges, and the other $45 - m$ pairs have passant join. Hence

$$\sum_i \binom{i}{2} n_i = 45 - m.$$

Subtracting half the first identity yields the decisive moment

$$4n_4 + 12n_6 + 24n_8 + \cdots = 10 - m. \tag{1}$$

No n_i with $i \geq 6$ can occur. The degree condition and (1) leave exactly

m	(n_2, n_4)	shape
6	$(33, 1)$	four isolated vertices on one four-point passant and six cycle vertices,
10	$(35, 0)$	ten cycle vertices; every passant meets S in 0 or 2.

Two degree-two vertices cannot by themselves form a cycle; in the first case uniqueness of the four-point passant puts all isolated vertices on that line.

The first shape has only four normalized cases. The effective stabilizer of a passant acts as D_{14} on its seven internal points, with four orbits on four-subsets:

representative	orbit size	$\lvert X\rvert$	secant degrees in X
0123	7	7	$(3, 3, 4, 4, 4, 4, 4)$
0124	14	3	$(0, 0, 0)$
0134	7	5	$(2, 2, 2, 4, 4)$
0135	7	5	$(2, 2, 2, 3, 3)$.

Here X is the set of internal points passantly joined to all four fixed points; retaining the three unused points of the fixed line only enlarges it. These four stabilizer orbits exhaust the four-subsets, and none of their induced secant graphs contains the required six-vertex two-regular subgraph. Thus the $m = 6$ shape is impossible.

For $m = 10$, fix $P \in S$. The remaining points consist of one point in each of the seven six-point passant fibres through P and an unordered pair among its 35 secant neighbors. The stabilizer $(G_0)_P \cong D_{28}$ has thirty-three orbits on the 595 pairs: five of size 7, sixteen of size 14, and twelve of size 28. For one representative of each orbit, add the seven fibre points in order, rejecting a prefix once a passant contains three selected points or a selected point has more than two secant neighbors. The complete certificate has last surviving-depth mass

depth	0	1	2	3	4
pair mass	28	35	91	238	203.

The masses sum to 595, and no representative survives the fifth fibre. Thus $m = 10$ is impossible as well.

Finally, the twelve internal points

$$\begin{gathered}(1:0:2),(1:3:2),(1:4:5),(1:1:8),(1:4:8),(1:1:7),\\(1:7:12),(1:3:3),(1:9:11),(1:10:11),(1:0:5),(1:8:7)\end{gathered}$$

have zero syndrome. Their stabilizer is dihedral of order 24. Therefore $d(K) = 12$.

3 Minimum geometry and weighted pairs

Fixed-point exhaustion first proves that there are 364 minimum supports in four orbits; the domain-size ledger may be skipped on a first reading. The matching model then identifies the four families, and their pair concurrences recover the incidence rows and the full elliptic scheme.

Fix one support point P. Its seven passant fibres again have six eligible points each, and it has 35 secant neighbors. Pencil parity leaves $s = 0, 2, 4$ secant neighbors. The corresponding multiplicities in the seven passant fibres, followed by the number of secant neighbors, are

$$(5, 1^6; 0), \qquad (3, 3, 1^5; 0), \qquad (3, 1^6; 2), \qquad (1^7; 4).$$

A syndrome-indexed exact check exhausts these four fixed-point domains. Its input sizes are respectively

$$7\binom{6}{5}6^6, \quad \binom{7}{2}\binom{6}{3}^2 6^5, \quad 7\binom{6}{3}6^6\binom{35}{2}, \quad 6^7\binom{35}{2}^2,$$

with disjoint secant pairs in the last domain. The certificate specifies the four partitions, proves their coverage, tests zero syndrome, and deduplicates only after conversion to unordered supports. The four domains contribute $0, 0, 0, 56$ words. The canonical replay generates the four projective PGL(2, 13)-orbits, and intersects each orbit with the supports containing P. These four fixed-point slices are disjoint, each has size 14, and their union is checked for equality with the exhaustive 56-word set. The order-28 stabilizer of the fixed point acts transitively on each slice; the stabilizer of a support inside it has order 2. Transitivity, or equivalently the count $56 \cdot 78/12 = 364$, proves global exhaustion.

The replay generates each orbit from its recorded representative and checks every support against M. The geometric constructions in Section 3.1 identify those orbits without representatives.

3.1 One octahedral and three toric families

Parametrize the conic by $(t^2 : t : 1)$, with its usual point at infinity. An internal point $P = (x : y : z)$ defines the fixed-point-free involution

$$j_P(t) = \frac{yt - x}{zt - y}, \qquad J_P = \begin{pmatrix} y & -x \\ z & -y \end{pmatrix}, \qquad J_P^2 = (y^2 - xz)I.$$

The seven secants through P are the seven pairs $\{t, j_P(t)\}$. Thus internal points may be read as perfect matchings of the fourteen conic points.

Fix a chord e, carry its endpoints to $\{0, \infty\}$, and let $N_e \cong D_{24}$ be their setwise stabilizer. The three supports over e are

$$S_r(e) = \{(x : 1 : r^{-1}x^{-1}) : x \in \mathbb{F}_{13}^\times\}, \qquad r \in \{2, 5, 11\}.$$

The second-conic upper-bound construction is due to Droms–Mellinger–Meyer [2, proof of Theorem 4.9]; here its $q = 13$ instances are separated and classified over each chord. Intrinsically, these three parameters satisfy $\chi(r) = -1$ and $\chi(r-1) = 1$. The support $S_r(e)$ is the punctured pencil conic

$$\{y^2 = rxz\}(\mathbb{F}_{13}) \setminus e,$$

so it has size twelve and stabilizer exactly N_e. At each displayed point, $\Delta = 1 - r^{-1} = (r-1)/r$ is a nonsquare; hence all twelve points are internal. The set is a codeword for a structural reason. If a passant $AX + BY + CZ = 0$ met $S_r(e)$ with odd multiplicity, the polynomial $Ax^2 + Bx + C/r$ would have a double root. Then

$$B^2 - 4AC = B^2(1 - r)$$

would be a square, contrary to the line being passant. Hence every passant meets $S_r(e)$ in zero or two points. There is no missing endpoint case: a passant contains neither endpoint of the chord $e = \{0, \infty\}$, so $A, C \neq 0$. Thus neither zero nor infinity is a root, and the displayed quadratic accounts for the entire intersection with the punctured conic.

For the remaining family, let $O \leq G_0$ be generated by $\left(\begin{smallmatrix} 1 & 0 \\ 3 & 8 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 4 \\ 3 & 0 \end{smallmatrix}\right)$. Exact multiplication gives orders $(4, 3, 2)$ for the generators and their product, and $|O| = 24$, so $O \cong S_4$, with conic orbits of sizes six and eight. On all 78 internal points, matching profile $(2, 3)$ inside those orbits selects a unique 12-point suborbit X_O (and two crossing pairs). The leaf checks all 78 passant parities and $\mathrm{Stab}_{G_0}(X_O) = O$, identifying the remaining minimum orbit.

All four stabilizers have order 24. Each family therefore has $2184/24 = 91 = \binom{14}{2}$ supports, agreeing with the fixed-point exhaustion above.

3.2 Exact arity-two reconstruction

We now forget how the supports were constructed and retain only their weighted pair graph. From that graph alone we recover first the six elliptic relations, then the incidence rows, and finally the code.

For internal points $P = (x_P : y_P : z_P)$ and Q, set

$$\Delta(P) = y_P^2 - x_P z_P, \qquad \beta(P, Q) = 2y_P y_Q - x_P z_Q - z_P x_Q,$$

and

$$\rho(P, Q) = \frac{\beta(P, Q)^2}{\Delta(P)\Delta(Q)}.$$

On distinct internal points, ρ takes the six values $0, 1, 3, 9, 10, 12$. Write A_r for the adjacency matrix of the non-diagonal relation $\rho(P, Q) = r$.

For distinct coordinates put

$$c(P, Q) = \#\{S \in \mathcal{H} : \{P, Q\} \subseteq S\}.$$

It suffices to count at one point. Fix $P = (1 : 0 : 2)$; one representative in each relation is displayed below.

ρ	0	1	3	9	10	12
Q	$(1:0:11)$	$(1:0:5)$	$(1:0:7)$	$(1:1:6)$	$(1:1:7)$	$(1:2:10)$
valency	7	14	14	14	14	14
$c(P,Q)$	8	6	6	12	7	9.

Direct counting of the four intrinsic support representatives gives these entries; transitivity on every relation transports them to all pairs. The four generated support orbits have size 91, are disjoint, and deduplicate to all 364 minimum supports. As a global check, the numbers of unordered pairs with concurrence $8, 6, 12, 7, 9$ are respectively $273, 1092, 546, 546, 546$. The exact replay is `verification/verify_pair_reconstruction.py`, with compact output in `verification/pair_reconstruction.json`. Only the relations A_1 and A_3 are fused. For a concurrence-six pair define

$$d_7(P, Q) = \#\{R : c(P, R) = c(R, Q) = 7\}.$$

The A_{10}^2 intersection row gives

$$d_7(P, Q) = 2 \quad \text{on } A_1, \qquad d_7(P, Q) = 4 \quad \text{on } A_3.$$

This pair-derived common-neighbor count recovers the full six-relation elliptic scheme.

Comparison with Hollmann–Xiang. Let u be their cross ratio, modulo inversion, and let $\lambda = f(u) = 1/4 + 1/(-2 + u + u^{-1})$ be their modified label. After swapping their first two conic coordinates and passing from points to polar lines, their cross-ratio formula [4, Theorem 5.3] gives

$$\lambda = \widehat{\rho}_{\mathrm{HX}}(P^\perp, Q^\perp) = 9\rho(P, Q).$$

Thus the source-to-paper dictionary is

$\{u, u^{-1}\}$	$\{12, 12\}$	$\{4, 10\}$	$\{3, 9\}$	$\{5, 8\}$	$\{2, 7\}$	$\{6, 11\}$
λ	0	9	1	3	12	4
paper relation	A_0	A_1	A_3	A_9	A_{10}	A_{12}.

The concurrence-eight relation recovers the incidence rows directly. Polarity sends $P = (x : y : z)$ to $P^\perp = (-z : 2y : -x)$, and

$$Q \in P^\perp \iff \beta(P, Q) = 0 \iff \rho(P, Q) = 0 \iff c(P, Q) = 8.$$

Consequently the seventy-eight concurrence-eight neighborhoods are exactly the seventy-eight rows of M.

The pair data also recovers the code without first naming a row. Let $(R_{\mathcal{H}})_{PQ} = c(P, Q) \bmod 2$ off the diagonal and $(R_{\mathcal{H}})_{PP} = 0$. Then

$$R_{\mathcal{H}} = A_{10} + A_{12}, \qquad \operatorname{rank} R_{\mathcal{H}} = 36, \qquad A_0 R_{\mathcal{H}} = 0,$$

so $\operatorname{im} R_{\mathcal{H}} = K$. Every coordinate occurs in $364 \cdot 12/78 = 56$ minimum supports, making unary data constant. Pair data suffices and unary data does not; reconstruction arity is exactly two. Indeed, intrinsic reconstruction is equivariant: the constant unary datum has automorphism group S_{78}, whereas Section 4 proves that the recovered marked presentation has automorphism group $\mathrm{PGL}(2,13)$. No intrinsic reconstruction of those nontrivial marked relations can therefore factor through unary data. Here "recovers" means the marked code–hypergraph presentation up to isomorphism. It does not choose an ordered projective frame or a labeling of the ground field.

4 The hidden field, spanning, and automorphisms

The recovered pair coloring now becomes an operator algebra. Its binary action projects onto K, the single operator A_9 supplies the scalar field $\mathbb{F}_8$, and the four orbit Grams become invertible scalars. The final anchor argument determines the full permutation group. For a first reading, the integral table may be skipped after retaining its binary action on K and the cubic satisfied by A_9.

These six level relations form the elliptic scheme. Polarity identifies A_0 with M, up to row order.

Only the following integral intersection rows are used:

	I	A_0	A_1	A_3	A_9	A_{10}	A_{12}
A_0^2	7	0	0	0	1	1	1
A_0A_9	0	2	2	2	0	2	0
A_0A_{10}	0	2	2	0	2	0	2
A_0A_{12}	0	2	0	2	0	2	2
A_9^2	14	0	4	2	2	1	4
A_{10}^2	14	0	2	4	2	4	1
A_{12}^2	14	4	2	2	3	2	2.

These numbers come from fixing $P = (1:0:2)$, using the six relation representatives in Section 3.2, and counting intermediate internal points. Transitivity on each ordered relation makes those seven representative rows the full matrix identities. Reducing modulo two gives

$$\begin{gathered} A_0^2 = I + A_9 + A_{10} + A_{12}, \\ A_0A_r = 0 \quad (r = 9, 10, 12), \\ A_1(A_9 + A_{10} + A_{12}) = 0, \qquad A_3(A_9 + A_{10} + A_{12}) = 0, \\ A_9^2 = A_{10}, \qquad A_{10}^2 = A_{12}, \qquad A_{12}^2 = A_9. \end{gathered} \tag{2}$$

These are parity reductions of intersection numbers. The $\mathrm{PGL}(2,13)$-action is transitive on each relation, so fixing one point and substituting one representative of every relation checks all entries.

Put $B = A_9$ and

$$e_K = I + A_0^2 = A_9 + A_{10} + A_{12}.$$

The same representative multiplication gives $e_K^2 = e_K$, while the two extra parity products in (2) give $A_1e_K = A_3e_K = 0$; also $A_0e_K = 0$. Hence $\operatorname{im} e_K \subseteq K$, while $e_Kx = x$ for every $x \in K$; consequently e_K has image K and rank 36. Thus on K, the seven Bose–Mesner basis elements act as

$$(I, A_0, A_1, A_3, A_9, A_{10}, A_{12}) = (1, 0, 0, 0, B, B^2, B^4).$$

Since $A_0B = 0$, the image of B lies in K. If $x \in K$ and $Bx = 0$, then (2) gives

$$0 = A_0^2x = (I + B + B^2 + B^4)x = x.$$

Thus B is injective on the 36-dimensional space K, so $\operatorname{im} B = K$, and B, B^2, B^4 all have rank 36.

Proposition 4.1 (The operator field)**.** *The action of the binary Bose–Mesner algebra on K has image $\mathbb{F}_8$. If α is a root of $t^3 + t^2 + 1$, then*

$$K \cong \mathbb{F}_8^{12}, \qquad (A_9, A_{10}, A_{12}) = (\alpha, \alpha^2, \alpha^4).$$

The identification of a named α is canonical up to Frobenius.

Exact elimination gives $\operatorname{rank}(B + I) = 78$. The replay `verification/verify_hidden_field.py` records independent bitset and dense eliminations in `verification/hidden_field.json`. Restricting the first identity in (2) to K and factoring

$$1 + t + t^2 + t^4 = (t + 1)(t^3 + t^2 + 1)$$

therefore gives $B^3 + B^2 + I = 0$. The cubic has no root in $\mathbb{F}_2$, so it is irreducible and makes K a vector space of dimension $36/3 = 12$ over $\mathbb{F}_8$. The squaring identities in (2) give the three displayed scalar operators. Together with the displayed action of all seven basis elements, this proves that the image of the whole binary Bose–Mesner algebra is exactly $\mathbb{F}_2[B] \cong \mathbb{F}_8$. This is an operator-field structure on the binary code, not a relabeling as a length-26 code over $\mathbb{F}_8$.

The module statement in the introduction follows by descent. By Madison–Wu [5, Theorem 5.1(i)], over $\overline{\mathbb{F}_2}$ the code is the direct sum of three non-isomorphic simple constituents of dimension twelve. Because K is twelve-dimensional over $\mathbb{F}_2[B]$, base change along its three embeddings in $\overline{\mathbb{F}_2}$ gives three $\mathrm{PSL}(2, 13)$-stable twelve-dimensional eigenspaces. These are precisely the Madison–Wu constituents. Frobenius cycles them transitively. A binary $\mathrm{PSL}(2, 13)$-submodule becomes, after scalar extension, a Frobenius-stable sum of a subset of these three non-isomorphic simples; the only such subsets are empty and full. Hence K is irreducible over $\mathbb{F}_2$. The constituents are absolutely simple and multiplicity-free, so endomorphism base change and Schur's lemma give

$$\operatorname{End}_{\mathbb{F}_2 \mathrm{PSL}(2,13)}(K) \otimes_{\mathbb{F}_2} \overline{\mathbb{F}_2} \cong \overline{\mathbb{F}_2}^{\,3};$$

the binary commutant therefore has dimension three over $\mathbb{F}_2$. This commutant contains $\mathbb{F}_2[B]$, so it equals $\mathbb{F}_8$. Since B also commutes with $\mathrm{PGL}(2, 13)$, inclusion of commutants gives

$$\operatorname{End}_{\mathbb{F}_2 \mathrm{PSL}(2,13)}(K) = \operatorname{End}_{\mathbb{F}_2 \mathrm{PGL}(2,13)}(K) = \mathbb{F}_8.$$

If N_i is the 91-by-78 support matrix of the i-th minimum-word orbit, every point lies on fourteen supports in that orbit. For one fixed point, the off-diagonal concurrence counts on $A_0, A_1, A_3, A_9, A_{10}, A_{12}$ are

$$\begin{array}{c|cccccc} N_1 & 0 & 2 & 2 & 5 & 0 & 2 \\ N_2 & 0 & 2 & 2 & 3 & 2 & 2 \\ N_3 & 4 & 2 & 0 & 2 & 4 & 1 \\ N_4 & 4 & 0 & 2 & 2 & 1 & 4. \end{array}$$

Each row is obtained from one intrinsic support representative and transported by transitivity of its 91-element orbit. Reducing these counts and the diagonal count fourteen modulo two gives

$$(N_1^{\mathsf{T}} N_1, N_2^{\mathsf{T}} N_2, N_3^{\mathsf{T}} N_3, N_4^{\mathsf{T}} N_4) = (A_9, A_9, A_{12}, A_{10}).$$

These are nonzero scalars on $\mathbb{F}_8^{12}$, hence automorphisms of K. Thus $\operatorname{im}(N_i^{\mathsf{T}} N_i) = K$ lies in the row space of N_i. Its rows already lie in K, so every orbit spans K.

To determine the scheme automorphisms, use the four anchors

$$P_0 = (1:0:2), \quad P_1 = (1:1:7), \quad P_2 = (1:0:7), \quad P_3 = (1:1:3).$$

The rigidity mechanism is summarized by

$$\underbrace{(P_0, P_1, P_2)}_{\text{(10,3,9) regular orbit}} \quad \Longrightarrow \quad \underbrace{P_3}_{\text{(3,1,9) is unique}} \quad \Longrightarrow \quad \underbrace{P \longmapsto \big([P = P_i], \rho(P, P_i)\big)_{i=0}^{3}}_{\text{injective on the 78 points}}.$$

The first three satisfy

$$\big(\rho(P_0, P_1), \rho(P_0, P_2), \rho(P_1, P_2)\big) = (10, 3, 9).$$

There are $78 \cdot 14 \cdot 2 = 2184$ ordered triples with this pattern: A_{10} has valency 14 and $p^{10}_{3,9} = 2$. Substitution in the symmetric-square action shows that the stabilizer of (P_0, P_1, P_2) in $\mathrm{PGL}(2, 13)$ is trivial. Since the group also has order 2184, it acts simply transitively on these triples.

For a fixed triple, P_3 is the unique point with relation signature $(3, 1, 9)$. The four values

$$\big(\rho(P, P_0), \rho(P, P_1), \rho(P, P_2), \rho(P, P_3)\big),$$

with equality to an anchor recorded on the diagonal, distinguish all 78 points. Consequently a scheme automorphism can be composed with a unique element of $\mathrm{PGL}(2, 13)$ to fix the first three anchors; it then fixes the fourth and every point. Thus the reconstructed scheme has automorphism group $\mathrm{PGL}(2, 13)$.

Every coordinate automorphism of K preserves $\mathcal{H}$. Conversely, $\mathcal{H}$ spans K, so every automorphism of $\mathcal{H}$ preserves K. The two coordinate-permutation groups coincide.

5 Recovery of the ambient conic plane

The last change of language recovers the plane from its internal-point scheme. The fourteen Sylow subgroups become the conic points, the involutions become the remaining points, and group-theoretic incidence becomes trace-polarity orthogonality in the adjoint model. Let

$$G = \mathrm{Aut}(\mathcal{H}) \cong \mathrm{PGL}(2, 13), \qquad \Omega = \mathrm{Syl}_{13}(G).$$

A Sylow-13 subgroup has normalizer of order 156, so $|\Omega| = 14$. Conjugation gives a sharply three-transitive action: in the standard model a Sylow subgroup is the unipotent group fixing one point of $\mathbf{P}^1(\mathbb{F}_{13})$, and both G and the set of ordered distinct triples have size 2184. Thus Ω is intrinsically the recovered conic-point set.

Let $\mathcal{I}$ be the set of involutions of G, and put $\Pi = \Omega \sqcup \mathcal{I}$. The adjoint model below will give $|\mathcal{I}| = 169$. Index both points and polar lines by Π. Define incidence by

(i) $U \in U^\perp$ for $U \in \Omega$, with no other Ω–Ω incidences;

(ii) $U \in j^\perp$ for $U \in \Omega$, $j \in \mathcal{I}$, exactly when j normalizes U, and symmetrically declare $j \in U^\perp$;

(iii) $i \in j^\perp$ for distinct $i, j \in \mathcal{I}$ exactly when ij is an involution.

These rules were defined intrinsically, before coordinates. The standard adjoint model is used only to prove that this incidence structure is the marked conic plane. Choose any isomorphism $G \to \mathrm{PGL}(2, 13)$ and transport the rules to that model; changing the choice merely composes the recovered plane with an automorphism and selects no preferred frame. Identify the standard plane with $\mathbf{P}(\mathfrak{sl}_2(\mathbb{F}_{13}))$. For traceless A, Cayley–Hamilton gives $A^2 = -\det(A)I$. The fourteen singular matrix lines are the nilpotent conic and correspond to the Sylow subgroups; the nonsingular lines correspond bijectively to projective involutions. Moreover

$$\langle A, B\rangle = \operatorname{tr}(AB)$$

is the polar form of the determinant conic. Orthogonality of two nilpotent lines means equality, orthogonality of a nilpotent and an involution means normalization of the Sylow subgroup, and for distinct involutions it is equivalent to their product being an involution. The three intrinsic rules therefore recover $\mathrm{PG}(2, 13)$, its conic Ω, and its polarity.

The 78 original coordinates are also intrinsic inside this plane. A coordinate stabilizer is a nonsplit-torus normalizer D_{28}. Its unique central involution has centralizer containing this D_{28}; the centralizer also has order 28, so the two groups are equal. The resulting equivariant map identifies the two coset G-sets and is therefore a bijection from the 78 coordinates to the involution class of centralizer order 28. The other involution class has size 91. Consequently the old matrix M is exactly the internal–internal block of the recovered polarity matrix. Section 3 reconstructed M and the elliptic scheme from weighted pairs, while Section 4 reconstructed their full automorphism group G. Hence the whole marked plane is reconstructed at arity two. This completes the proof of Theorem 1.1.

6 Proof and verification boundary

The human proof owns the reductions: tangent products produce the local weight-eight graph, line moments produce the two weight-ten shapes, conic pencils produce the toric words, weighted pairs produce the polarity rows, and the adjoint representation produces the ambient plane. Exact execution enters only after those reductions.

The accompanying `verification/` directory contains separate evidence artifacts and replay records for the theta matrix, moment–stabilizer leaves, weighted-pair reconstruction, toric–octahedral geometry, ambient plane, and hidden operator field. The earlier clique and syndrome programs remain as independent checks but are not theorem-facing proof leaves. The minimum-word exhaustion and four-anchor rigidity retain their original exact replays. The entry point `verify_evidence.py` checks byte counts and SHA-256 hashes before running every program. All arithmetic is exact and deterministic. From the paper root, the public command

```
make check
```

ends with `Paper-IV q=13 evidence: PASS` and then builds the warning-free PDF. The exact programs are described in `verification/README.md`; the formal package is distributed separately and carries its own README. The Python evidence uses only the standard library, the manuscript build uses a Nix-pinned TeX environment, and the formal package a Nix-pinned Lean environment.

The shared Lean library formalizes the normalized conic and incidence code, the theta inequality, moment compression, pair-neighborhood interface, pencil discriminant identity, and hidden-cubic cancellation. The paper-owned package checks pair-only recovery, unary constancy, the three toric parities, the hidden cubic on the relation-operator image, and both projective-plane uniqueness axioms on the normalized 183-point model. Table 1 records the remaining division of proof.

claim	proof mode	exact boundary
dimension 36	published theorem; Lean theorem	Madison–Wu gives the formula; exact elimination and semantic recovery maps prove rank 42 at $q = 13$.
weight 8	human/classical reduction; exact PSD certificate; Lean implication	Segre's lemma gives $h = 1$; the saturated pencils and three cyclic 14-orbits give the 42-vertex graph. Exact Fourier traces and rational Schur elimination certify the rank-28 form; Lean proves the quadratic-form clique implication.
weight 10	human moment proof; finite certificates; Lean transport	Incidence and pair counts give (1); four D_{14} and thirty-three D_{28} representatives close the two shapes.
minimum layer	Lean native evaluation; trusted Python; human transport	Both executions exhaust the 56 fixed-point words; projective transitivity and the double count give all 364.
minimum geometry	human conic proof; finite certificate; Lean parity checks	The toric families have a discriminant proof. One octahedral matching profile and the four stabilizers are checked in standard form.
pair reconstruction	human polarity proof; finite certificate; Lean theorem	Pair counts and one common-neighbor count recover the scheme; color-eight neighborhoods recover the rows, and pair parity recovers the code.
automorphism group	human concurrence transport; Lean finite classification; trusted replay	Human transport makes every support automorphism preserve the six-valued scheme; the formal finite leaves identify its automorphisms with the 2184 projective maps, and Python independently replays the four-anchor test.
ambient plane	human adjoint proof; finite certificate; Lean incidence gate	The Sylow/involution rules are identified with trace polarity. Lean checks the 183-point plane axioms; the group-theoretic census is exact Python.
hidden field and spanning	human algebra; finite certificate; Lean theorem	The irreducible cubic makes $K \cong \mathbb{F}_8^{12}$; checked Gram scalars and a kernel-checked image argument prove that every orbit spans.

Table 1: Proof modes and trust boundaries for Theorem 1.1.

The paper-owned aggregate namespace is `PassantCodeQ13.Gates.Main`; its terminals are

```
incidenceRankAndCodeDimension
fixedPointWeightTwelveExhaustion
minimumOrbitsSpanRhoZeroKernel
ellipticSchemeAutomorphismsAreProjective
pairOnlyReconstruction
toricMinimumSupports
hiddenFieldCubicOnImage
ambientPlaneIncidence.
```

The shared low-weight namespace is `RelativeConicArcs.Gates.PassantCodeQ13`; its terminals are

```
weightEight_semantic_transport
arbitrary_weightTen_profile_transport.
```

Native evaluation compiles and runs finite decision procedures rather than reducing them entirely in the kernel. Under Lean 4.32.0-rc1, the tracked `#print axioms` audit therefore exposes each such use as a declaration-local axiom; the symbolic transport theorems remain kernel-checked relative to those finite leaves. Thus "Lean theorem" in Table 1 means kernel-checked relative to its printed axioms, whereas "Lean native evaluation" and "trusted Python" are executions. Hash checks identify the executed sources and outputs but do not turn either execution into a kernel-checked certificate; no finite leaf in the table is described that way. The formal scope retains three explicit human transports. Projective transitivity and double counting carry the fixed-point exhaustion to all 364 minimum words. Weighted-pair equivariance carries support automorphisms to the polar-relation scheme. The adjoint trace model identifies the abstract Sylow/involution incidence with the conic polarity. Segre's lemma of tangents remains the cited classical input to the weight-eight transport.

7 Conclusion

The minimum words retain the geometry that row reduction forgets. Their weighted pairs recover the parity checks and code, then the elliptic scheme, and finally the full conic plane and polarity. The theta form and global line moment explain why this reconstruction first appears in weight twelve; the toric–octahedral families explain what appears there; and the hidden $\mathbb{F}_8$ orbit-Gram scalars directly certify that each family spans. Thus, in this $q = 13$ instance, the weighted minimum-support 2-section, passant incidence matrix, binary code, elliptic scheme, and marked conic plane are mutually recoverable up to the stated isomorphisms.

The principal point is an exact information threshold. Unary statistics of the minimum layer carry no distinguishing information, whereas its weighted 2-section reconstructs the marked code and then the complete ambient conic plane. Pair data are therefore a complete invariant of the marked presentation up to isomorphism.

Clebsch: Rigidity from Sparse Shadows

I From deep holes, a form arises;
II beneath it, paired chords turn true;
III through paired veils, the two sheets exchange;
IV **from bare, whispered words, a plane rises;**
V beneath one form, two shadows, a hidden twist between.

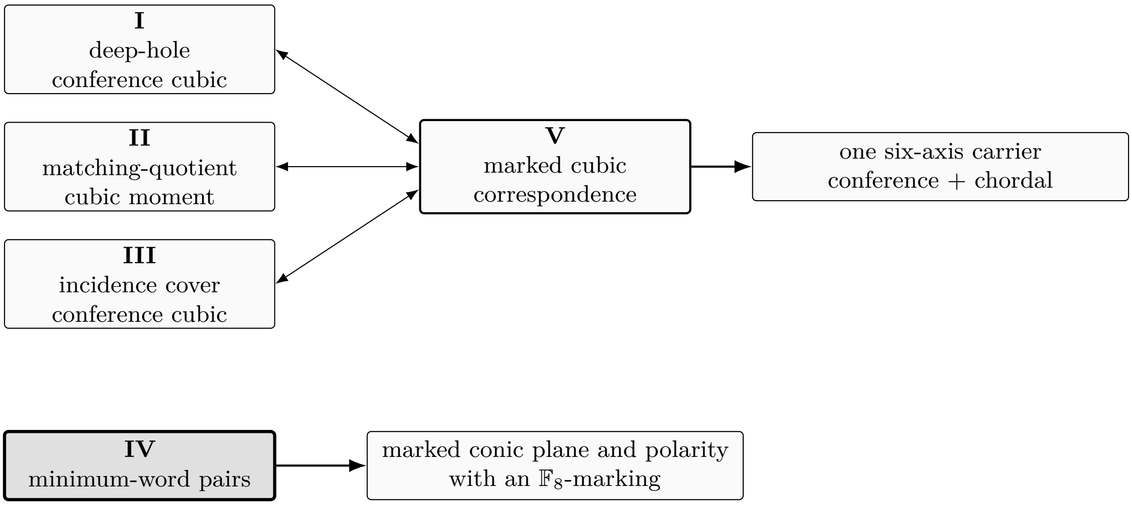


Figure 1: Programme map. The upper double arrows are Paper V's marked transports and returns between the chordal and conference shadows; the right arrow is its common-carrier reconstruction. The lower solid arrow is Paper IV's independent minimum-word reconstruction. Only Paper V's global-negation/$\mathbb{F}_4$ torsor and Paper IV's $\mathbb{F}_8$ Frobenius-orbit marking are compared across the two branches; the chordal-line torsor is not.

The five papers are logically independent; the map records reconstruction correspondences and thematic relations, not proof dependencies.

Here the sparse shadow is literal: the 364 minimum words of the binary conic code are the starting data, and their weighted pair concurrences are the only input the reconstruction uses.

This paper's place in the programme is methodological: minimum-word pairs give a separate instance in which sparse data recover a substantially richer marked source.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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